

Problem 15.23

You are given the following information:

- (i) The present value of a $6n$ -year annuity-immediate of 1 at the end of every year is 9.7578.
- (ii) The present value of a $6n$ -year annuity-immediate of 1 at the end of every second year is 4.760.
- (iii) The present value of a $6n$ -year annuity-immediate of 1 at the end of every third year is K .

Determine K assuming an annual effective interest rate of i .

Solution

Let $v = (1 + i)^{-1}$. From the first annuity,

$$9.7578 = \frac{1 - v^{6n}}{i}.$$

From the second annuity, payments are made at times $2, 4, \dots, 6n$, so

$$4.760 = \frac{1 - v^{6n}}{(1 + i)^2 - 1}.$$

Dividing the first equation by the second gives

$$\frac{9.7578}{4.760} = \frac{(1 + i)^2 - 1}{i}.$$

Since

$$\frac{(1 + i)^2 - 1}{i} = \frac{2i + i^2}{i} = 2 + i,$$

we have

$$i = \frac{9.7578}{4.760} - 2 \approx 0.04995798.$$

Now use the first equation to find

$$1 - v^{6n} = 9.7578i.$$

Thus,

$$1 - v^{6n} \approx 9.7578(0.04995798) \approx 0.4874800.$$

For payments at the end of every third year, the present value is

$$K = \frac{1 - v^{6n}}{(1 + i)^3 - 1}.$$

Therefore,

$$\begin{aligned} K &= \frac{0.4874800}{(1.04995798)^3 - 1} \\ &\approx 3.0954. \end{aligned}$$

Hence,

$$\boxed{K \approx 3.0954}.$$